Statistics

Exam revision through worked illustrations

Dispersion • Skewness • Kurtosis
Covariance • Pearson and rank correlation

Purpose. Learn Unit 3 through clear theory followed by boxed illustrations. Each box gives an exam-style problem first, followed by its detailed solution: formula, substitutions, working, answer, interpretation and fx-991CW instructions.

What to study. Read the problem, cover its solution and attempt every requested part before checking the working. Use the calculator to shorten the arithmetic while keeping the written method clear.

Correlation illustrations. Given values appear in tables before the solution. Further tables show deviations, products or paired ranks, so you can follow exactly where every total comes from.

How to use these 30 pages

1. **Understand the method.** Read the theory and its illustration together, including the topics listed at the top of the box.
2. **Practise the calculator route.** "Yes, directly" means read a result. "Calculator arithmetic" means type the expression. "Manual setup first" means choose classes or ranks yourself.
3. **Revise the essentials.** Use the two formula sheets on pages 29–30, then revisit any illustration whose setup you cannot reproduce.

Exam point. Yes, you can obtain standard deviation directly on the fx-991CW. Choose σ_x for a population or s_x for a sample. The notes beneath the formulas identify comparable shortcuts for the other topics.

This edition follows the supplied Unit 3 notes and syllabus image, with the same explanation style, tables, boxed illustrations and calculator guidance. It focuses on the concepts and their worked illustrations.

READ THIS BEFORE REVISING

A short route through the unit

Part	What you should be able to do	Pages
Foundations	Identify population/sample; find mean, median and mode	3–4
Variance and SD	Use deviations; distinguish N from $n-1$	5–6
Positions and range	Find quartiles, IQR, QD and percentiles	7–8
Frequency/grouped data	Use counts, midpoints and interpolation	9–10
Using SD	Apply Chebyshev, the empirical rule and z-scores	11–12
Skewness	Read tails, compare centres and quartile gaps	13–14
Skewness coefficients	Use Pearson, Bowley and Kelly	15–16
Moments and kurtosis	Calculate central moments and classify kurtosis	17–18
Covariance / Pearson	Read paired tables and standardize joint variation	19–20
Correlation methods	Use product tables and Spearman ranks, including ties	21–23
Relative variation / growth	Compare CV; average compound change	24
Calculator workflows	Enter raw, frequency and paired data efficiently	25–26
Exam explanations / visuals	Explain interpretation, causation and graph choice	27–28
Formula sheets	Recall formulas and the matching calculator route	29–30

Conventions that prevent different answers

Population versus sample. An entire population uses N, μ, σ. A sample used to estimate population variation uses $n, \bar{x}, s$ and variance denominator $n-1$. The calculator gives both options; the question decides which to select.

Raw quartiles. Sort, find the overall median, and take the medians of the lower and upper halves. Exclude the overall median from both halves for an odd-sized list. Compare calculator quartiles with the rule your teacher requires.

Other percentiles. Here $L = np$: an integer position means average observations $L, L+1$; otherwise round the position up. A prescribed classroom rule takes priority. Grouped quantiles use class interpolation instead.

Rank ties. Equal values receive average ranks. Exact Spearman correlation is Pearson correlation of those ranks. Use the short rank-difference formula only without ties.

Rounding and units. Keep full calculator precision until the final answer. Variance has squared units, SD has original units, and correlation is unitless.

Exam point. Every illustration begins with a "Topics covered" line. In the correlation parts, read the given-data table first and then follow the working table. Preserve the row pairing throughout.

FOUNDATIONS

What are we trying to describe?

A data set is a collection of observations, such as the weights of 24 tomatoes. **Descriptive statistics** summarize the observations in front of us. **Inferential statistics** use a sample to say something about a larger population.

The average grade of one class describes that class. Using it to estimate the average grade of all ten classes is inference. A **population parameter** describes the whole population; a **sample statistic** is computed from the selected sample.

Illustration 1

Topics covered: Population; sample; parameter; statistic; descriptive versus inferential statistics.

Problem. A class has 40 students and a mean mark of 68. Five selected students score 60, 65, 70, 75 and 80. (a) Identify the population and sample. (b) Find the sample mean. (c) Identify the parameter and statistic. (d) Distinguish describing these five students from estimating the class mean.

Solution.

1. Identify the groups. The population is all 40 students, so $N = 40$. The five selected students are the sample, so $n = 5$.

2. Calculate the sample mean.

$$\bar{x} = \frac{\sum x}{n} = \frac{60 + 65 + 70 + 75 + 80}{5} = \frac{350}{5} = \boxed{70}.$$

3. Name the quantities. The full-class mean $\mu = 68$ is a population parameter. The calculated $\bar{x} = 70$ is a sample statistic.

4. Interpret. Reporting 70 as the mean of these five students is descriptive. Using 70 to estimate the entire class mean is inference. Here it differs from the known population mean by 2 marks.

fx-991CW – Yes, directly. Enter the five scores in 1-Variable; read $n = 5$ and $\bar{x} = 70$. The calculator computes the mean; you identify its population or sample interpretation.

Three measures of the centre

Measure	Meaning	How to obtain it
Mean	Share the total equally among observations	Add all values and divide by their number
Median	Middle of the ordered observations	Sort; use the middle value or average the two middle values
Mode	Most frequent value	Count how often each value occurs

For raw data, x means a data value and $\sum$ means "add all the indicated terms".

$$\mu = \frac{\sum x}{N}, \qquad \bar{x} = \frac{\sum x}{n}.$$

fx-991CW – Direct result. Statistics → 1-Variable → TOOLS → 1-Var Results: read $\bar{x}$. It gives the mean directly for either interpretation.

Here N is population size, n is sample size, μ is population mean and $\bar{x}$ is sample mean. The arithmetic is the same; the interpretation differs.

Illustration 2

Topics covered: Arithmetic mean; median; mode; sorting; effect of an extreme value.

Problem. The scores of five students are 2, 3, 3, 4 and 8. Find their mean, median and mode. If the score 8 is replaced by 28, recalculate all three measures and explain which is most affected.

Solution.

1. Original centre. The ordered scores are $2, 3, 3, 4, 8$, with total 20 and $n = 5$.

$$\bar{x} = \frac{20}{5} = \boxed{4}, \qquad \text{Median} = x_{(3)} = \boxed{3}, \qquad \text{Mode} = \boxed{3}.$$

The median is the third observation because two lie on each side. The mode is 3 because it occurs twice.

2. Changed score. Replacing 8 by 28 increases the total by 20, to 40. The order becomes $2, 3, 3, 4, 28$.

$$\bar{x}_{\text{new}} = 40/5 = \boxed{8}, \quad \text{Median}_{\text{new}} = \boxed{3}, \quad \text{Mode}_{\text{new}} = \boxed{3}.$$

3. Conclusion. The mean rises by 4; the median and mode do not change. The mean is most affected by this extreme observation.

fx-991CW – Direct + manual. Read mean and Med in 1-Var Results for each list. Identify the most frequent value manually; there is no dedicated mode result.

The four questions in Unit 3

1. **Centre:** what is a typical value? Use mean, median or mode.
2. **Spread:** how far apart are the values? Use range, IQR, variance or SD.
3. **Shape:** is one tail longer, and how pronounced are extreme values? Use skewness and kurtosis.
4. **Association:** do two measured quantities change together? Use covariance, a scatterplot or correlation.

Illustration 3

Topics covered: Mean; range; interpretation of consistency; raw/frequency/paired calculator input.

Problem. Two delivery teams have times A: 9, 10, 11 minutes and B: 1, 10, 19 minutes. Compare their means and ranges and identify the more consistent team. If each time is also paired with a delivery distance, which calculator data type should be used to study their association?

Solution.

1. Compare centres.

$$\bar{x}_A = (9 + 10 + 11)/3 = \boxed{10}, \qquad \bar{x}_B = (1 + 10 + 19)/3 = \boxed{10}.$$

2. Compare spreads. Range is largest minus smallest:

$$R_A = 11 - 9 = \boxed{2 \text{ min}}, \qquad R_B = 19 - 1 = \boxed{18 \text{ min}}.$$

3. Interpret. A is more consistent despite the equal means. A centre alone cannot show this difference in spread.

4. Choose the data format. One list of times uses 1-Variable. Counts of repeated times use Frequency On. Matched distance–time observations require 2-Variable, keeping the measurements from each delivery together.

fx-991CW – Direct + arithmetic. Read each mean directly; read max/min and subtract for range. Use paired entry for association, not independently sorted columns.

Exam point. Start every numerical by identifying the data format: raw list, frequency table, class intervals, or paired observations. Then identify whether a population or sample formula is required.

FOUNDATIONS / VARIANCE AND STANDARD DEVIATION

Why subtract, square, divide and take a root?

A **deviation** is an observation minus the mean. Above-mean values have positive deviations; below-mean values have negative deviations. Their total is zero, so simply averaging signed deviations cannot measure spread.

Square the deviations to prevent cancellation. For an entire population, average those squares to obtain variance, then take the positive square root to return to the original units:

$$\mu = \frac{\sum x}{N}, \qquad \sigma^2 = \frac{\sum (x-\mu)^2}{N}, \qquad \sigma = \sqrt{\sigma^2}.$$

N is the number of population observations, x an individual observation, μ the population mean, σ^2 its variance and σ its SD. Variance has squared units; SD has the original units. SD is a root-mean-square deviation, not a mean absolute deviation.

fx-991CW – Yes, directly. Enter the raw observations in 1-Variable and read σ_x^2 for population variance and σ_x for population SD. These are direct outputs.

Illustration 4

Topics covered: Mean; deviations; squared deviations; population variance; population SD; units.

Problem. The lengths of every component in a batch are 2, 4, 6 and 8 cm. Calculate the population mean, variance and standard deviation using a deviation table. State the units of each result.

Solution.

1. Select the population formula. All components in the stated batch are included, so $N = 4$.

$$\mu = \frac{\sum x}{N} = \frac{2+4+6+8}{4} = \boxed{5 \text{ cm}}.$$

2. Subtract the mean, then square.

x	2	4	6	8
$x-5$	-3	-1	1	3
$(x-5)^2$	9	1	1	9

The ordinary deviations sum to zero; their squares sum to $9+1+1+9=20$.

3. Divide by N and take the positive root.

$$\sigma^2 = \frac{\sum (x-\mu)^2}{N} = \frac{20}{4} = \boxed{5 \text{ cm}^2}, \quad \sigma = \sqrt{5} = \boxed{2.23607 \text{ cm}}.$$

Variance has squared units; SD returns to the original centimetres.

fx-991CW – Yes, directly. Enter the four lengths; read $\bar{x} = 5$, $\sigma_x^2 = 5$ and $\sigma_x = 2.23607$. SD is a direct result; no manual deviation table is needed on the calculator.

Exam point. Use the population denominator because the question includes every member of the defined batch. A larger SD means greater spread around the centre; it does not necessarily mean a larger mean.

VARIANCE / SAMPLE AND COMPUTATIONAL METHOD

Choose the denominator before calculating

A sample is usually used to estimate variation in a larger population. Its fitted mean $\bar{x}$ uses one degree of freedom, leaving $n-1$ independent deviations. The usual sample formulas are

$$s^2 = \frac{\sum(x-\bar{x})^2}{n-1}, \qquad s = \sqrt{s^2}.$$

Here n is the number of sample observations. Bessel's correction makes s^2 an unbiased estimator of population variance under standard random sampling; it does not make s exactly unbiased.

The following identity avoids a long deviation table:

$$\sum(x-\bar{x})^2 = \sum x^2 - \frac{(\sum x)^2}{n}.$$

$\sum x^2$ adds individual squares; $(\sum x)^2$ squares the total. Compute this total first, then divide by $n-1$ for sample variance or by N when describing an entire population.

fx-991CW – Yes, directly. Read s_x^2, s_x for sample variance/SD. The same Statistics results provide $\sum x$ and $\sum x^2$ if you want to verify the written computational method.

Illustration 5

Topics covered: Sample mean; sample variance and SD; Bessel correction; degrees of freedom.

Problem. A sample of ten daily boat-production counts is 17, 21, 18, 27, 17, 21, 20, 22, 18 and 23. Find the sample mean, variance and SD. A manager considers SD above 3 boats/day unacceptable. Should she be concerned?

Solution.

1. **Identify the denominator.** These are sample days, so $n = 10$ and variance divides by $n-1 = 9$.
2. **Calculate totals and the mean.**

$$\sum x = 204, \quad \sum x^2 = 4250, \qquad \bar{x} = \frac{204}{10} = \boxed{20.4 \text{ boats/day}}.$$

3. **Obtain the squared-deviation total.**

$$\sum(x-\bar{x})^2 = \sum x^2 - \frac{(\sum x)^2}{n} = 4250 - \frac{204^2}{10} = 88.4.$$

4. **Calculate sample spread.**

$$s^2 = \frac{88.4}{9} = \boxed{9.82222 \text{ (boats/day)}^2}, \qquad s = \sqrt{88.4/9} = \boxed{3.13404 \text{ boats/day}}.$$

5. **Decide.** Yes: $3.13404 > 3$. Do not round to 3 before comparing. The population SD would be $\sqrt{88.4/10} = 2.97321$, but it is not the requested sample SD. There are 9 degrees of freedom.

fx-991CW – Yes, directly. Enter the ten counts. Check $n = 10, \bar{x} = 20.4$; read s_x^2 and s_x, not the σ outputs.

Exam point. Write the selected formula and substitution even when the calculator gives the result directly. A sample mean still divides by n; only the usual sample variance uses $n-1$.

FOUNDATIONS / POSITION AND DISTANCE

Range, quartiles and interfractile range

Range uses only the two extremes:

$$R = x_{\max} - x_{\min}.$$

fx-991CW – Calculator arithmetic. Read $\max(x)$ and $\min(x)$ in 1-Var Results, then subtract. There is no separate range result.

$x_{\max}$ is the largest observation and $x_{\min}$ the smallest. Range ignores the remaining values; the actual extremes must be known.

Illustration 6

Topics covered: Maximum; minimum; range; sensitivity to an outlier; open-ended classes.

Problem. The weights of four parcels are 4, 5, 5 and 6 kg. Find the range. Recalculate it if the last weight becomes 60 kg. Can an exact range be obtained from the class labels "below 5", "5–10" and "above 10" alone? Explain.

Solution.

1. Original range. Minimum is 4 and maximum is 6:

$$R = x_{\max} - x_{\min} = 6 - 4 = \boxed{2 \text{ kg}}.$$

2. Changed range. The new maximum is 60; the minimum remains 4:

$$R_{\text{new}} = 60 - 4 = \boxed{56 \text{ kg}}.$$

Only one observation changed, yet the range rose by 54 kg. It depends only on the two extremes and is sensitive to an outlier.

3. Open classes. No exact range can be found: "below 5" does not reveal the smallest weight and "above 10" does not reveal the largest.

fx-991CW – Calculator arithmetic. Read $\max(x)$ and $\min(x)$, then subtract. When both extremes are already given, type the subtraction directly. Missing extremes cannot be recovered by the calculator.

Quartiles: divide the ordered data into quarters

Q_1, Q_2, Q_3 mark approximately the lower quarter, median and upper quarter. Sort first; find the overall median. Then take the medians of the lower and upper halves, excluding the overall median from both halves when the count is odd. Finite-list percentages need not be exact.

For the middle-half spread and its half-width,

$$\text{IQR} = Q_3 - Q_1, \qquad \text{QD} = \frac{Q_3 - Q_1}{2}.$$

fx-991CW – Direct + arithmetic. Read Med and quartiles directly if their convention matches the question; subtract for IQR and divide by 2 for QD.

Illustration 7

Topics covered: Ordered positions; even/odd medians; lower/upper quartiles; IQR; quartile deviation.

Problem. Find the median, lower quartile, upper quartile, IQR and quartile deviation for 253, 104, 633, 57, 500, 201, 43, 380, 467, 162, 220 and 302. Use the median-of-halves rule.

Solution.

1. Sort before selecting positions.

43	57	104	162	201	220
253	302	380	467	500	633

Illustration 7 (continued)

2. Find the middle of the complete list. There are $n = 12$ values, so average positions 6 and 7:

$$Q_2 = \frac{x_{(6)} + x_{(7)}}{2} = \frac{220 + 253}{2} = \boxed{236.5}.$$

3. Find each half's median. The lower half occupies positions 1–6; the upper half occupies 7–12:

$$Q_1 = \frac{104 + 162}{2} = \boxed{133}, \qquad Q_3 = \frac{380 + 467}{2} = \boxed{423.5}.$$

4. Subtract, then halve.

$$\text{IQR} = 423.5 - 133 = \boxed{290.5}, \qquad \text{QD} = 290.5/2 = \boxed{145.25}.$$

The middle half spans 133 to 423.5. For an odd-sized list, exclude the overall median before taking half-list medians.

fx-991CW – Direct + arithmetic. Enter the 12 values and read Med, Q_1, Q_3 if the required convention matches. Subtract for IQR and halve for QD.

Fractiles, percentiles and deciles

A p-fractile is a cutoff associated with fraction p below it. Percentiles use hundredths; deciles use tenths, so $D_8 = P_{80}$. An interfractile range is the difference between two chosen cutoffs, such as $P_{90} - P_{10}$. Use the explicitly stated position rule in the problem below.

Illustration 8

Topics covered: Fractile; percentile; decile; integer/fractional positions; interfractile range.

Problem. The following ordered quiz scores are given. Find P_{80}, the eighth decile D_8, P_{82}, and the central-80% interfractile range. Use $L = np$: average positions $L, L + 1$ when L is an integer; otherwise round L up.

59	65	67	68	71	72	75	79	81	83
85	87	88	91	92	93	94	95	100	100

Solution.

1. Find the 80th-percentile position. Here $n = 20$ and $p = 0.80$, so $L = 20(0.80) = 16$. Average the 16th and 17th observations:

$$P_{80} = \frac{93 + 94}{2} = \boxed{93.5}, \qquad D_8 = P_{80} = \boxed{93.5}.$$

2. Handle a fractional position. For P_{82}, $L = 20(0.82) = 16.4$. Round up to position 17, so $P_{82} = x_{(17)} = \boxed{94}$.

3. Find the lower/upper 10% cutoffs.

$$P_{10} = \frac{x_{(2)} + x_{(3)}}{2} = \frac{65 + 67}{2} = 66, \quad P_{90} = \frac{x_{(18)} + x_{(19)}}{2} = \frac{95 + 100}{2} = 97.5.$$

$$P_{90} - P_{10} = 97.5 - 66 = \boxed{31.5 \text{ marks}}.$$

fx-991CW – Manual positions first. Type np, locate the ordered values yourself, then average/subtract. General percentiles are not direct Statistics outputs.

Exam point. State the percentile convention. A position tells you where to look; a percentile is the actual value found there.

FOUNDATIONS / FREQUENCIES AND CLASS INTERVALS

Use every observation represented by a frequency

A frequency f is a count: if $x = 2$ occurs three times, it contributes $3(2)$ to the total and $3(2^2)$ to the sum of squares. The sample size is $n = \sum f$, not the number of distinct values.

$$\bar{x} = \frac{\sum fx}{n}, \qquad s^2 = \frac{\sum fx^2 - (\sum fx)^2/n}{n-1}, \qquad s = \sqrt{s^2}.$$

For a complete population use $N = \sum f$ and $\sigma^2 = \sum fx^2/N - \mu^2$. Every repeated observation contributes, so frequency weights every term.

fx-991CW – Yes, directly. Set Frequency On before entering values and counts. Read mean, population/sample variance and SD directly; the CW applies the counts automatically.

Illustration 9

Topics covered: Frequency counts; weighted mean; computational sum of squares; sample/population variance and SD.

Problem. A sample of couples reports the preferred number of cars per family as shown. Find the mean, sample variance and sample SD. State the population SD if these responses are instead treated as the full population.

Cars x	0	0.5	1	1.5	2	2.5
Frequency f	2	14	23	7	4	2

Solution.

1. Form weighted totals.

x	f	fx	fx^2
0	2	0	0
0.5	14	7	3.5
1	23	23	23
1.5	7	10.5	15.75
2	4	8	16
2.5	2	5	12.5
Total	52	53.5	70.75

2. Find the mean and squared-deviation total.

$$\bar{x} = \frac{53.5}{52} = \boxed{1.02885 \text{ cars}}, \qquad \sum f(x-\bar{x})^2 = 70.75 - \frac{53.5^2}{52} = 15.70673077.$$

3. Divide by the sample degrees of freedom and take the root.

$$s^2 = \frac{15.70673077}{51} = \boxed{0.30798 \text{ cars}^2}, \qquad s = \boxed{0.55496 \text{ cars}}.$$

4. Change the interpretation. If the 52 responses are the population, $\sigma^2 = 15.70673077/52 = 0.30205$ and $\sigma = \boxed{0.54959 \text{ cars}}$. The mean is unchanged. Half-car answers are averages of partners' preferences, not physical half-cars.

fx-991CW – Yes, directly. Enter the six values and frequencies; verify $n = 52$. Read $\bar{x}$, s_x^2, s_x for the sample; use σ_x for the alternative population interpretation.

Exam point. Do not confuse $fx^2 = f \times x^2$ with $(fx)^2$. In Statistics, displayed sums already include the frequency weights.

GROUPED DATA / MIDPOINTS AND INTERPOLATION

From class intervals to estimates

For class limits, midpoint $m = (\text{lower}+\text{upper})/2$. Use midpoints and class frequencies for approximate mean and SD. Grouping loses individual positions inside each class.

For a grouped quantile, form cumulative counts and find the class containing target count pN:

$$Q_p = L + \frac{pN - C}{f}h.$$

Here L is its lower boundary, C the cumulative count before it, f its frequency and h its width. Use $p = 0.25, 0.5, 0.75$ for the quartiles. For equal-width classes, the mode is

$$\text{Mode} = L + \frac{f_1 - f_0}{2f_1 - f_0 - f_2}h,$$

with f_1 the modal-class count and f_0, f_2 its neighbours. Integer classes 40–49, 50–59 use boundaries 39.5, 49.5, 59.5; continuous classes use their given endpoints.

Illustration 10

Topics covered: Midpoints; frequencies; grouped mean/variance/SD; cumulative counts; quartiles; median; mode.

Problem. The following continuous grouped data represent a population. Estimate its mean, variance, SD, three quartiles and mode.

Class	0–10	10–20	20–30	30–40	40–50
Frequency	5	9	12	8	6

Solution.

1. Find midpoints and totals. The classes mean $0 \le x < 10$, $10 \le x < 20$, etc.

m	f	Cum. f	fm	fm^2
5	5	5	25	125
15	9	14	135	2025
25	12	26	300	7500
35	8	34	280	9800
45	6	40	270	12150
	40		1010	31600

2. Compute the midpoint-based estimates.

$$\mu \approx 1010/40 = \boxed{25.25}, \quad \sigma^2 \approx 31600/40 - 25.25^2 = \boxed{152.4375}, \quad \sigma \approx \boxed{12.34656}.$$

3. Locate target counts $10, 20, 30$ in their own classes.

$$Q_1 \approx 10 + \frac{10-5}{9}(10) = \boxed{15.55556}, \qquad Q_2 \approx 20 + \frac{20-14}{12}(10) = \boxed{25},$$

$$Q_3 \approx 30 + \frac{30-26}{8}(10) = \boxed{35}.$$

4. Use the largest count, 12, for the modal class 20–30.

$$\text{Mode} \approx 20 + \frac{12-9}{24-9-8}(10) = \boxed{24.28571}.$$

fx-991CW – Direct + manual setup. Enter midpoints/counts for mean and population variance/SD. Select each quantile/modal class yourself and type the interpolation formula; midpoint-list quartiles are not grouped interpolations.

FOUNDATIONS / USING THE STANDARD DEVIATION

Chebyshev, the empirical rule and z-scores

Chebyshev's theorem: no bell shape required

For a distribution with a finite mean μ and positive finite SD σ, the proportion within k standard deviations of the mean is at least

$$1 - \frac{1}{k^2}, \qquad k > 1.$$

fx-991CW – Calculator arithmetic. Type $(1 - 1/k^2) \times 100$ for the guaranteed percentage, and $\mu - k\sigma$, $\mu + k\sigma$ for endpoints. No Chebyshev menu is needed.

The corresponding interval is $[\mu - k\sigma, \mu + k\sigma]$. Here k is a positive *number of standard deviations*, not a data value. This gives a minimum proportion, not the exact proportion.

Illustration 11

Topics covered: Distribution-free lower bound; k standard deviations; endpoints; percentages; minimum counts.

Problem. A population of 200 observations has mean 100 and SD 10. Without assuming normality, find intervals within 2 and 3 SDs and the minimum number of observations guaranteed in each. Explain whether these counts are exact.

Solution.

1. Two-SD interval and proportion.

$$\mu \pm 2\sigma = 100 \pm 20 = [80, 120], \qquad 1 - 1/2^2 = 0.75.$$

At least $200(0.75) = \boxed{150}$ observations lie in $[80, 120]$.

2. Three-SD interval and proportion.

$$\mu \pm 3\sigma = 100 \pm 30 = [70, 130], \qquad 1 - 1/3^2 = 8/9.$$

The bound is $200(8/9) = 177.777\ldots$. Since a count is an integer, at least $\boxed{178}$ observations lie in $[70, 130]$.

3. Interpret. These are lower bounds, not exact counts. Chebyshev needs $k > 1$ and a positive finite SD, but no bell-shape assumption.

fx-991CW – Calculator arithmetic. Calculate the endpoints and $200(1 - 1/k^2)$. Round a fractional minimum count upward, not to the nearest integer.

Empirical rule: approximately normal data

For an approximately normal, bell-shaped distribution, about 68%, 95%, and 99.7% lie within 1, 2 and 3 SDs respectively. Calculate the corresponding endpoints as $\mu \pm k\sigma$.

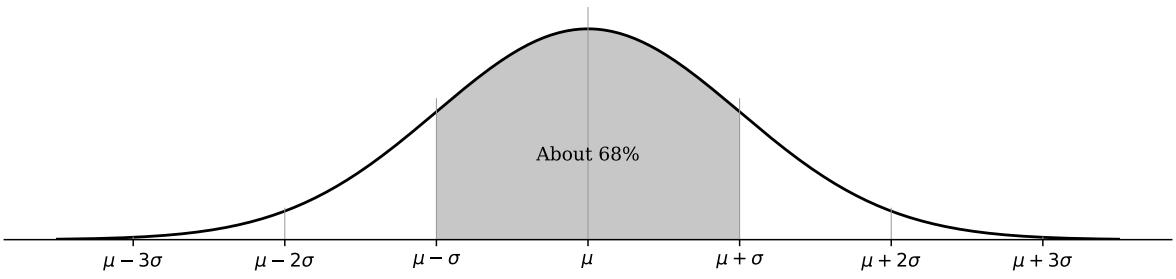


Illustration 12

Topics covered: Normal-shape assumption; 68–95–99.7 rule; standard-deviation intervals; contrast with Chebyshev.

Problem. One thousand approximately normal measurements have mean 50 and SD 5. Use the empirical rule to estimate how many lie within 1, 2 and 3 SDs. If normality were not given, what minimum count could Chebyshev guarantee within 2 SDs?

Solution.

1. Form each interval as $\mu \pm k\sigma$.

Interval	Empirical proportion	Estimated count
$50 \pm 5 = [45, 55]$	68%	$1000(0.68) = 680$
$50 \pm 10 = [40, 60]$	95%	$1000(0.95) = 950$
$50 \pm 15 = [35, 65]$	99.7%	$1000(0.997) = 997$

2. Change the assumption. Without normality, Chebyshev gives $1 - 1/2^2 = 0.75$, so at least $1000(0.75) = \boxed{750}$ values are in $[40, 60]$.

3. Distinguish the conclusions. The normal-rule counts are approximate estimates. The Chebyshev count is a guaranteed minimum under its conditions.

fx-991CW – Calculator arithmetic. Remember the percentages, then calculate endpoints and population size × proportion. The calculator cannot justify normality from these summaries alone.

A z-score describes one observation

$$z = \frac{x - \mu}{\sigma}.$$

fx-991CW – Calculator arithmetic. Type $(x - \mu) \div \sigma$ in Calculate, using the mean and SD given or obtained from Statistics. You need not find a special z-score button.

x is the observed value, μ the population mean, σ the population SD, and z the number of SDs above or below the mean. A positive z means above the mean, a negative z below, and zero at the mean.

To reverse the calculation, use $x = \mu + z\sigma$. The corresponding descriptive sample standardization is $(x - \bar{x})/s$; it is not automatically a hypothesis-test statistic.

Illustration 13

Topics covered: Positive/negative/zero z; units; inverse standardization; sample versus population summaries.

Problem. A population has mean 50 and SD 5. Find and interpret the z-scores of observations 60, 40 and 50. Find the original observation corresponding to $z = 1.5$.

Solution.

1. Standardize each observation using $z = (x - \mu)/\sigma$.

$$z_{60} = (60 - 50)/5 = \boxed{2}, \quad z_{40} = (40 - 50)/5 = \boxed{-2}, \quad z_{50} = (50 - 50)/5 = \boxed{0}.$$

Thus 60 is two SDs above the mean, 40 is two below, and 50 is at the mean.

2. Reverse the standardization. Rearranging $z = (x - \mu)/\sigma$ gives $x = \mu + z\sigma$:

$$x = 50 + 1.5(5) = \boxed{57.5}.$$

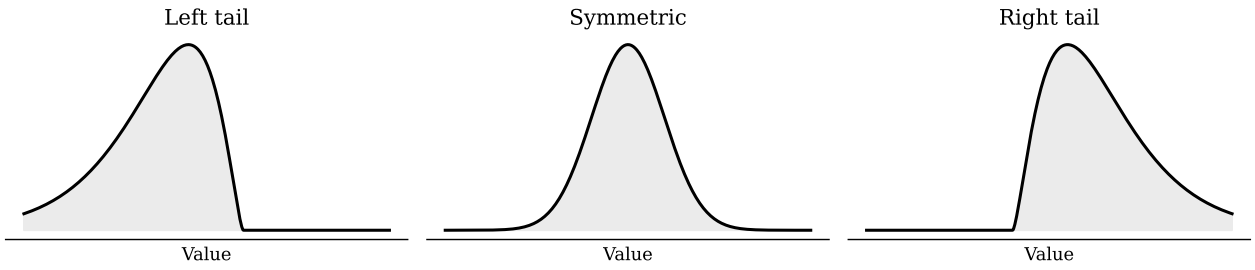
The z-scores are unitless; 57.5 has the original data units. If using sample summaries descriptively, use $(x - \bar{x})/s$ consistently.

fx-991CW – Calculator arithmetic. Type $(60 - 50)/5$, $(40 - 50)/5$, $(50 - 50)/5$ and $50 + 1.5 \times 5$. Obtain the correct mean/SD directly from Statistics if raw data are given.

SHAPE / SKEWNESS

Which side has the longer tail?

Skewness describes asymmetry. Imagine the main cluster of observations and then look at the tail extending away from it. The *tail*, not the side with the tallest bars, gives the direction.



Shape	Tail direction	Usual ordering in a simple one-peak distribution
Negative skew	Longer left tail	Mean $<$ median $<$ mode
Symmetric	Balanced tails	Mean $=$ median $=$ mode, when one central mode exists
Positive skew	Longer right tail	Mode $<$ median $<$ mean

Illustration 14

Topics covered: Positive/negative/symmetric shapes; tail direction; mean–median–mode; outlier influence.

Problem. For each distribution below, calculate the mean, median and mode, then identify whether it is positively skewed, negatively skewed or symmetric. Explain your decision.

A	$0, 1, 1, 2, 3, 4, 10$
B	$-10, -4, -3, -2, -1, -1, 0$
C	$-3, -2, -1, 0, 0, 0, 1, 2, 3$

Solution.

1. Compute centres. Use mean $= \sum x/n$; read the middle ordered position and count repetitions.

	n	$\sum x$	Mean	Median	Mode
A	7	21	$21/7 = 3$	$x_{(4)} = 2$	1
B	7	-21	$-21/7 = -3$	$x_{(4)} = -2$	-1
C	9	0	$0/9 = 0$	$x_{(5)} = 0$	0

2. Identify the shape. A has a longer right tail and Mode $<$ Median $<$ Mean: positive skew. B has the reflected left tail and Mean $<$ Median $<$ Mode: negative skew. C is balanced about zero and is symmetric. These lists illustrate the usual ordering; that ordering is not a theorem for every distribution.
fx-991CW – Direct + manual. Enter each list and read mean/Med directly. Count modes and assess tail direction yourself; the sign of the raw mean is not a skewness test.

Why does the mean move toward the tail?

The mean uses every value, so an extreme observation pulls it toward the tail. The median uses position. Symmetry does not require a single peak or a normal distribution.

Graphical and numerical evidence

- A histogram with a longer right tail suggests positive skewness; a longer left tail suggests negative skewness.
- Compare the gaps $Q_3 - Q_2$ and $Q_2 - Q_1$. A larger upper gap gives positive Bowley skewness; a larger lower gap gives negative Bowley skewness.
- Compare mean and a meaningful mode, or mean and median, using the coefficient specified in the question.

A symmetric graph need not be normal. Equal quartile gaps, or a zero skewness coefficient, need not prove the entire distribution is symmetric; different measures summarize different aspects of asymmetry.

Illustration 15

Topics covered: Graphical versus numerical evidence; quartile gaps; asymmetry; limits of a summary.

Problem. A distribution has $Q_1 = 10$, median $Q_2 = 20$ and $Q_3 = 40$. Compare its two median-to-quartile gaps, calculate Bowley's coefficient, and state whether these results alone prove the shape of the entire distribution.

Solution.

1. Calculate both gaps. Lower gap $= Q_2 - Q_1 = 20 - 10 = 10$; upper gap $= Q_3 - Q_2 = 40 - 20 = 20$.

2. Standardize their difference.

$$\mathrm{Sk}_B = \frac{(Q_3 - Q_2) - (Q_2 - Q_1)}{Q_3 - Q_1} = \frac{20 - 10}{40 - 10} = \boxed{\frac{1}{3}}.$$

3. Interpret the evidence. The positive value says the upper middle gap is larger. A histogram with a longer right tail would support positive skewness. However, three quartiles do not fully describe the outer tails, so this result alone does not prove the full distribution's shape.

fx-991CW – Calculator arithmetic. Type $(40 + 10 - 2 \times 20)/(40 - 10)$. The coefficient is an expression to calculate, not a direct Statistics result.

Absolute measure or relative coefficient?

An **absolute measure**, such as mean minus mode, has the original units. A **relative coefficient** divides by a spread measure so that units cancel. Relative coefficients can be compared across numeric scales more meaningfully.

Illustration 16

Topics covered: Absolute skewness; relative coefficient; units; changing measurement scale.

Problem. Lengths measured in centimetres have mean 60, mode 54 and SD 12. Find absolute skewness and Pearson's coefficient. Convert all lengths to millimetres, recalculate both measures, and explain what changes.

Solution.

1. Work in centimetres.

$$\text{Absolute skewness} = \mu - \text{Mode} = 60 - 54 = \boxed{6 \text{ cm}},$$

$$\mathrm{Sk}_P = \frac{\mu - \text{Mode}}{\sigma} = \frac{6}{12} = \boxed{0.5}.$$

2. Convert consistently. Multiply mean, mode and SD by 10: $600, 540, 120$ mm.

$$\text{Absolute skewness} = 600 - 540 = \boxed{60 \text{ mm}}, \quad \mathrm{Sk}_P = \frac{600 - 540}{120} = \boxed{0.5}.$$

3. Explain. The absolute measure changes numerical value with units. The unitless relative coefficient stays unchanged because both its numerator and denominator scale by the same factor.

fx-991CW – Calculator arithmetic. Type the subtraction and divide by the matching SD. If raw lengths are given, read mean/SD directly, but identify the numeric mode manually.

SHAPE / COEFFICIENTS OF SKEWNESS

Choose the formula from the given quantities

1. Karl Pearson: use a clear mode when available

$$\mathrm{Sk}_P = \frac{\bar{x} - \mathrm{Mode}}{s}.$$

fx-991CW – Calculator arithmetic. No direct Pearson-skewness output. Read mean and SD from Statistics, determine the mode, then calculate (mean − mode)÷SD.

$\bar{x}$ is the mean, Mode the most frequent numeric value, and s the SD used for that data set. For population data substitute μ, σ. If mode is not clear, use Pearson's second coefficient:

$$\mathrm{Sk}_{P,2} = \frac{3(\bar{x} - \mathrm{Median})}{s}.$$

fx-991CW – Calculator arithmetic. Read mean, Med and the appropriate SD in Statistics; evaluate 3(mean − Med) ÷ SD. This is not a separate result-menu item.

It is a separate coefficient, motivated by the approximate empirical relationship Mode $\approx 3\mathrm{Median} - 2\bar{x}$; the two coefficients need not be identical.

Illustration 17

Topics covered: Pearson first/second coefficients; when mode is unclear; factor 3; sign; absolute measures.

Problem. Distribution A has mean 40, mode 34 and SD 12. Distribution B has mean 40, median 38 and SD 12, with no clear mode. Find an appropriate Pearson skewness coefficient for each and interpret the signs.

Solution.

1. A has a usable mode. Apply Pearson's first coefficient:

$$\mathrm{Sk}_{P,A} = \frac{\mathrm{Mean} - \mathrm{Mode}}{\mathrm{SD}} = \frac{40 - 34}{12} = \boxed{0.5}.$$

2. B has a median but no clear mode. Apply Pearson's second coefficient, including its factor 3:

$$\mathrm{Sk}_{P,2,B} = \frac{3(\mathrm{Mean} - \mathrm{Median})}{\mathrm{SD}} = \frac{3(40 - 38)}{12} = \boxed{0.5}.$$

3. Interpret. Both coefficients indicate positive skewness. Their numerators are 6 in the original units; the final coefficients are unitless. Equal coefficients here do not imply identical distributions or guarantee that the two formulas always agree.

fx-991CW – Calculator arithmetic. Type $(40 - 34)/12$ and $3(40 - 38)/12$. No direct skewness output exists; obtain mean/Med/SD directly when raw data are supplied.

2. Bowley: use the quartiles

$$\mathrm{Sk}_B = \frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1} = \frac{(Q_3 - Q_2) - (Q_2 - Q_1)}{Q_3 - Q_1}.$$

fx-991CW – Calculator arithmetic. Read or calculate the required quartiles first; type $(Q_3 + Q_1 - 2Q_2) \div (Q_3 - Q_1)$. There is no Bowley button.

Q_1, Q_2, Q_3 are lower quartile, median and upper quartile.

Illustration 18

Topics covered: Lower/upper quartiles; median; absolute quartile asymmetry; Bowley coefficient and sign.

Problem. The lower quartile, median and upper quartile are 12, 20 and 36. Calculate absolute quartile asymmetry, the IQR and Bowley's coefficient of skewness. Interpret the result.

Solution.

1. **Compare the gaps.** Lower gap $= 20 - 12 = 8$; upper gap $= 36 - 20 = 16$.
2. **Calculate absolute asymmetry and spread.**

$$\text{Absolute quartile asymmetry} = 16 - 8 = \boxed{8}, \qquad \text{IQR} = 36 - 12 = \boxed{24}.$$

3. **Divide the gap difference by the total gap.**

$$\text{Sk}_B = \frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1} = \frac{36 + 12 - 2(20)}{36 - 12} = \frac{8}{24} = \boxed{\frac{1}{3}}.$$

The coefficient is positive: the upper part of the middle half spreads farther from the median. It describes middle-half asymmetry, not every feature of the tails.

fx-991CW – Calculator arithmetic. Read or calculate the required quartiles first; type $(36+12-2\times20)/(36-12)$.

3. Kelly: use percentiles or deciles

$$\text{Sk}_K = \frac{P_{90} + P_{10} - 2P_{50}}{P_{90} - P_{10}} = \frac{D_9 + D_1 - 2D_5}{D_9 - D_1}.$$

fx-991CW – Manual setup first. Find P_{10}, P_{50}, P_{90} by the required percentile method, then substitute. The CW does not supply general percentiles directly in its Statistics results.

Illustration 19

Topics covered: Percentiles; deciles; central 80%; Kelly skewness; sign and denominator.

Problem. A distribution has $P_{10} = 8$, $P_{50} = 20$ and $P_{90} = 44$. Find Kelly's coefficient of skewness and the central-80% interfractile range. State the equivalent decile notation.

Solution.

1. **Match percentiles to deciles.** $D_1 = P_{10} = 8$, $D_5 = P_{50} = 20$, $D_9 = P_{90} = 44$.
2. **Calculate the coefficient.**

$$\text{Sk}_K = \frac{P_{90} + P_{10} - 2P_{50}}{P_{90} - P_{10}} = \frac{44 + 8 - 2(20)}{44 - 8} = \frac{12}{36} = \boxed{\frac{1}{3}}.$$

3. **Find the interfractile spread.**

$$P_{90} - P_{10} = 44 - 8 = \boxed{36}.$$

The upper median gap is $44 - 20 = 24$, larger than the lower gap $20 - 8 = 12$, so the positive sign is consistent. The coefficient would be undefined if its denominator were zero.

fx-991CW – Calculator arithmetic. Type $(44 + 8 - 2 \times 20)/(44 - 8)$ and $44 - 8$. If percentiles must first be found from raw data, apply the specified percentile rule manually.

Exam point. The sign gives direction. Bowley and Kelly lie in $[-1, 1]$ when defined; Pearson need not. All coefficients are undefined when their denominator is zero. Numeric skewness is not meaningful for nominal labels such as colours.

SHAPE / KURTOSIS AND MOMENTS

Symmetry and tail weight are different questions

Kurtosis describes the importance of extreme deviations from the mean. The slides introduce it as "peakedness". That is common introductory exam language, but peak height alone is not a reliable definition: the fourth-power formula is especially sensitive to extreme observations.

Central moments: averages of powers of deviations

For a full data distribution with values x, frequencies f and $N = \sum f$, define

$$\mu = \frac{\sum fx}{N}, \qquad m_2 = \frac{\sum f(x-\mu)^2}{N}, \quad m_3 = \frac{\sum f(x-\mu)^3}{N}, \quad m_4 = \frac{\sum f(x-\mu)^4}{N}.$$

fx-991CW – Direct + prepared data. Mean and $m_2 = \sigma_x^2$ are direct results. For m_3, m_4, form cubed/fourth-power deviations, sum with frequencies, then divide by N.

m_2, m_3, m_4 mean the average squared, cubed and fourth-power deviations respectively. These descriptive moments divide by N; do not substitute the sample variance with denominator $N-1$ into these formulas.

Moment	Meaning
m_2	Variance of the empirical distribution; measures spread
m_3	Signed cubed deviations; supports a skewness measure
m_4	Fourth-power deviations; gives large weight to extremes

Illustration 20

Topics covered: Frequency-weighted mean; second/third/fourth central moments; even/odd powers.

Problem. For the population frequency table below, calculate the mean and the second, third and fourth central moments. Explain why the third moment vanishes.

Value x	-1	0	1
Frequency f	1	2	1

Solution.

1. Find the centre. $N = \sum f = 4$ and

$$\mu = \frac{\sum fx}{N} = \frac{1(-1) + 2(0) + 1(1)}{4} = \boxed{0}.$$

2. Raise deviations to each power. Since $\mu = 0$, $d = x - \mu = x$.

x	f	fd^2	fd^3	fd^4
-1	1	1	-1	1
0	2	0	0	0
1	1	1	1	1
Total	4	2	0	2

3. Divide by the population count.

$$m_2 = 2/4 = \boxed{0.5}, \quad m_3 = 0/4 = \boxed{0}, \quad m_4 = 2/4 = \boxed{0.5}.$$

Equal positive and negative cubed deviations cancel. Even powers do not cancel.
fx-991CW – Direct + prepared powers. With frequencies entered, read mean 0 and $\sigma_x^2 = m_2 = 0.5$ directly. Sum cubed/fourth-power deviations and divide by 4 for m_3, m_4.

Pearson's moment coefficients

$$\gamma_1 = \frac{m_3}{m_2\sqrt{m_2}}, \qquad \beta_1 = \frac{m_3^2}{m_2^3} = \gamma_1^2, \qquad \beta_2 = \frac{m_4}{m_2^2}, \qquad \gamma_2 = \beta_2 - 3.$$

fx-991CW – Calculator arithmetic. No direct moment-skewness or kurtosis output. Once m_2, m_3, m_4 are known, type these ratios and subtract 3 for excess kurtosis.

γ_1 is signed moment skewness; β_1 loses the sign. β_2 is kurtosis and γ_2 is **excess** kurtosis. These complete the moment coefficients named in the slides. Software may use bias-adjusted sample versions; the convention here is stated explicitly.

Type	β_2	Excess γ_2
Leptokurtic	> 3	> 0
Mesokurtic	$= 3$	$= 0$
Platykurtic	< 3	< 0

The normal distribution has $\beta_2 = 3$. Thus $\beta_2 = 4.2$ means leptokurtic and excess $= 1.2$; $\beta_2 = 2$ means platykurtic and excess $= -1$.

Illustration 21

Topics covered: Signed moment skewness; beta-one; Pearson kurtosis; excess kurtosis; all three classifications.

Problem. The central moments of a distribution are $m_2 = 4$, $m_3 = -4$ and $m_4 = 64$. Find γ_1, β_1, Pearson kurtosis β_2 and excess kurtosis γ_2. State the direction of skewness and classify kurtosis.

Solution.

1. Calculate signed moment skewness.

$$\gamma_1 = \frac{m_3}{m_2\sqrt{m_2}} = \frac{-4}{4\sqrt{4}} = \frac{-4}{8} = \boxed{-0.5}.$$

2. Square it for β_1. $\beta_1 = \gamma_1^2 = (-0.5)^2 = \boxed{0.25}$. Squaring removes the sign, so use γ_1 for direction.

3. Standardize the fourth moment and find excess.

$$\beta_2 = \frac{m_4}{m_2^2} = \frac{64}{16} = \boxed{4}, \qquad \gamma_2 = \beta_2 - 3 = 4 - 3 = \boxed{1}.$$

4. Classify. Negative γ_1 indicates negative moment skewness. Since $\beta_2 > 3$ (equivalently excess > 0), the distribution is leptokurtic. Mesokurtic uses $\beta_2 = 3$; platykurtic uses $\beta_2 < 3$.

fx-991CW – Calculator arithmetic. Type $-4/(4^{1.5})$, square the result, then calculate $64/(4^2)$ and subtract 3. These coefficients are not direct Statistics outputs.

Scope note. The final slide also names a Kelly kurtosis measure but supplies neither a formula nor a convention. No unsupported Kelly-kurtosis numerical is imposed here. Learn its name from the slide; use Pearson's stated moment formula for the worked calculations unless your instructor supplies a different formula.

Exam point. Do not compare excess kurtosis with 3. Compare β_2 with 3 or excess γ_2 with 0. A distribution can be symmetric and still leptokurtic. The moment illustrations above show how to calculate these quantities.

SYLLABUS SUPPLEMENT / COVARIANCE AND CORRELATION

Do two quantities move together?

Source: Syllabus image; the attached slide decks do not develop this topic.

Paired data contain two measurements on the *same* unit: for example, study hours x and examination mark y for one student. Each row is one pair. Reordering the x column independently of y destroys the relationship.

Sample covariance measures joint movement

$$s_{xy} = \frac{\sum(x-\bar{x})(y-\bar{y})}{n-1}.$$

fx-991CW – Calculator arithmetic. No separate covariance result. In 2-Variable, obtain $n, \sum x, \sum y, \sum xy$ and evaluate $[\sum xy - (\sum x)(\sum y)/n]/(n-1)$.

n is the number of pairs, $\bar{x}, \bar{y}$ are their sample means, and s_{xy} is sample covariance. The numerator adds products of deviations.

- Both above their means or both below: the product is positive.
- One above and the other below: the product is negative.
- A positive total suggests the quantities tend to move together; a negative total suggests opposite movement.

Covariance units multiply: hours × marks, for example. Its magnitude changes if hours become minutes, so magnitude alone is hard to compare across scales.

Illustration 22

Topics covered: Pairing; sample means; deviation products; covariance sign and units; population versus sample.

Problem. Calculate the sample covariance of study hours and marks from the following matched observations. State its sign and units, and give the population covariance if these three students form the whole population.

Student	1	2	3
Hours x	1	2	3
Marks y	2	4	3

Solution.

1. Calculate means. $n = 3$, $\bar{x} = (1+2+3)/3 = 2$, $\bar{y} = (2+4+3)/3 = 3$.

2. Calculate paired deviation products.

Student	$x-2$	$y-3$	$(x-2)(y-3)$
1	−1	−1	1
2	0	1	0
3	1	0	0
Total	0	0	1

3. Divide by the correct count.

$$s_{xy} = \frac{\sum(x-\bar{x})(y-\bar{y})}{n-1} = \frac{1}{2} = \boxed{0.5 \text{ hours} \times \text{marks}}.$$

The sign is positive. If all three students form the population, divide by 3 instead: $\boxed{1/3 \text{ hours} \times \text{marks}}$.

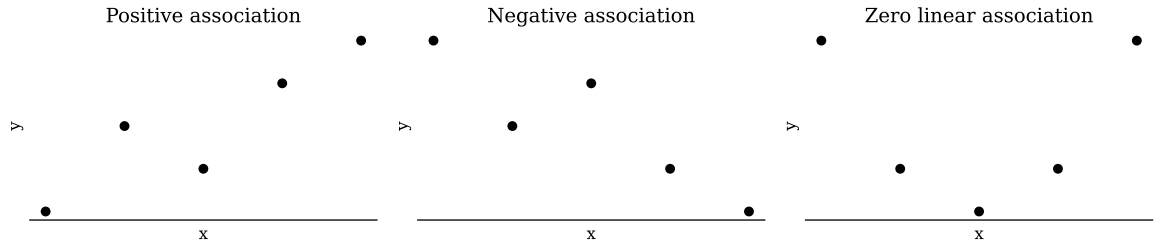
fx-991CW – Sums, then arithmetic. Enter original pairs; obtain $n = 3, \sum x = 6, \sum y = 9, \sum xy = 19$. Type $(19 - 6 \times 9/3)/(3-1) = 0.5$. Covariance has no separate result entry.

Pearson correlation removes the units

$$r = \frac{n\sum xy - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}.$$

fx-991CW – Yes, directly. Enter the original pairs in Statistics → 2-Variable; choose TOOLS → Reg Results → y=a+bx and read r. No manual product table is needed to obtain r.

n counts pairs; $\sum xy$ adds row products. Correlation is unitless, from -1 to 1; it is undefined if either variable is constant.



r near $+1$ means a strong positive linear association; near -1 means a strong negative linear association. $r = 0$ means no linear association. The U-shaped example above has $r = 0$ even though y clearly depends on x.

Illustration 23

Topics covered: Standardizing covariance; Pearson r; sign/strength; constant variables; zero versus nonlinear association.

Problem. (a) Find Pearson's correlation for Table A. (b) For Table B, show that zero Pearson correlation does not rule out a relationship.

Table A: x	1	2	3
Table A: y	2	4	3

Table B: x	-2	-1	0	1	2
Table B: y	4	1	0	1	4

Solution.

1. Table A: form the five totals. There are $n = 3$ matched pairs.

x	y	x^2	y^2	xy
1	2	1	4	2
2	4	4	16	8
3	3	9	9	9
6	9	14	29	19

2. Substitute the totals.

$$r_A = \frac{3(19) - 6(9)}{\sqrt{[3(14) - 6^2][3(29) - 9^2]}} = \frac{3}{\sqrt{6(6)}} = \boxed{0.5}.$$

This is positive, imperfect linear association. The coefficient is unitless.

3. Table B: check the product sum. $n = 5$, $\sum x = 0$, $\sum y = 10$, $\sum x^2 = 10$, $\sum y^2 = 34$ and $\sum xy = 0$. Thus $r_B = [5(0) - 0(10)]/\sqrt{[5(10) - 0^2][5(34) - 10^2]} = \boxed{0}$. Yet every pair satisfies $y = x^2$, a U-shaped relationship. Zero correlation means no linear association, not no relationship.

fx-991CW – Yes, directly. Enter each original paired table in 2-Variable; linear Reg Results gives 0.5 and 0 respectively. A constant variable would instead give an undefined coefficient.

Exam point. Make a scatterplot first when possible. Pearson measures a straight-line pattern and can be strongly affected by an outlier. A large correlation does not establish causation.

CORRELATION METHODS / PRODUCT TABLE

Pearson: a table that avoids repeated subtraction

For n matched pairs, make columns x, y, x^2, y^2, xy and add each column. Use the totals directly:

$$r = \frac{n\sum xy - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}.$$

$\sum x^2$: square each value, then add. $(\sum x)^2$: square the column total. Sample covariance:

$$s_{xy} = \frac{\sum xy - (\sum x)(\sum y)/n}{n-1}.$$

fx-991CW – Direct + arithmetic. Covariance needs arithmetic. Pearson r is direct: 2-Variable → TOOLS → Reg Results → y=a+bx.

Illustration 24

Topics covered: Given paired table; raw product totals; means; direct substitution; sample covariance; Pearson correlation.

Problem. The heights and weights of nine people are given below. Calculate Pearson's correlation coefficient and sample covariance. Interpret the coefficient.

Height x (in)	57	59	62	63	64	65	55	58	57
Weight y (lb)	113	117	126	126	130	129	111	116	112

Solution.

1. Extend the paired table. Here $n = 9$. Keep each person's measurements together.

x	y	x^2	y^2	xy
57	113	3249	12769	6441
59	117	3481	13689	6903
62	126	3844	15876	7812
63	126	3969	15876	7938
64	130	4096	16900	8320
65	129	4225	16641	8385
55	111	3025	12321	6105
58	116	3364	13456	6728
57	112	3249	12544	6384
540	1080	32502	130072	65016

2. Substitute into Pearson's formula. Use $n = 9$ and the five totals above:

$$r = \frac{9(65016) - 540(1080)}{\sqrt{[9(32502) - 540^2][9(130072) - 1080^2]}}$$

$$= \frac{1944}{\sqrt{918(4248)}} = \boxed{0.98443}.$$

3. Calculate sample covariance. The means are $540/9 = 60$ and $1080/9 = 120$.

$$s_{xy} = \frac{\sum xy - (\sum x)(\sum y)/n}{n-1} = \frac{65016 - 540(1080)/9}{9-1} = \boxed{27 \text{ in} \times \text{lb}}.$$

There is very strong positive linear association between height and weight in this sample. It is not proof of causation.

fx-991CW – Yes, directly for r. Enter the original paired data in 2-Variable, check $n = 9$, means 60 and 120, then read linear $r = 0.98442507$. For covariance calculate 216/8.

CORRELATION METHODS / RANKS WITHOUT TIES

Spearman: compare the order of the observations

Spearman correlation measures association in ordering. Assign rank 1 to the smallest value in each variable, preserving the original paired rows. Let R, T be the two ranks and $d = R - T$.

When there are no ties and n is the number of pairs,

$$\rho_s = 1 - \frac{6\sum d^2}{n(n^2-1)}.$$

fx-991CW – Manual ranks, direct result. Rank both columns yourself. Enter the paired ranks in 2-Variable and read linear r; it is Spearman for the original observations. You can also type the displayed no-ties formula.

Illustration 25

Topics covered: Given paired table; ranking direction; preserved rows; rank differences; no-ties Spearman; interpretation.

Problem. Calculate Spearman's rank correlation for the following observations. Show the ranks and squared rank differences, and interpret the result.

x	10	20	30	40	50
y	30	10	20	50	40

Solution.

1. Rank smallest as 1 in both variables. The first $x = 10$ is smallest (rank 1), while its paired $y = 30$ is third-smallest (rank 3). Keep the original rows.

Pair	x	y	R	T	$d = R - T$	d^2
1	10	30	1	3	−2	4
2	20	10	2	1	1	1
3	30	20	3	2	1	1
4	40	50	4	5	−1	1
5	50	40	5	4	1	1
					Total	8

2. Use the no-ties formula. There are $n = 5$ pairs and $\sum d^2 = 8$.

$$\rho_s = 1 - \frac{6\sum d^2}{n(n^2-1)} = 1 - \frac{6(8)}{5(25-1)} = 1 - \frac{48}{120} = \boxed{0.6}.$$

3. Interpret. The rankings have positive, imperfect agreement. They are not identical: several observations occupy different positions.

fx-991CW – Manual ranks, direct result. Enter the paired R, T columns as x, y in 2-Variable; read linear $r = 0.6$. Original values would give Pearson rather than this rank coefficient.

Exam point. Use the same ranking direction in both columns. Sorting the columns separately and leaving them sorted would destroy the original pairs and manufacture agreement.

CORRELATION METHODS / TIED RANKS

Spearman when values are repeated

Ties receive average ranks. If two equal values occupy positions 2 and 3, each receives $(2+3)/2 = 2.5$. The exact Spearman coefficient is Pearson correlation of these average ranks:

$$\rho_s = \frac{n\sum RT - (\sum R)(\sum T)}{\sqrt{[n\sum R^2 - (\sum R)^2][n\sum T^2 - (\sum T)^2]}}.$$

R, T are the two ranks in each row; n counts pairs. This is the same Pearson formula with ranks replacing x, y. The no-ties shortcut is not generally exact with repeated ranks.

fx-991CW – Manual ranks, direct result. Create the average ranks first. Enter those paired ranks in 2-Variable and read linear r; no separate tie-correction command is needed.

Illustration 26

Topics covered: Given paired table; average ranks; products and squares of ranks; exact Spearman; calculator entry.

Problem. Find Spearman's rank correlation between the following mothers' and daughters' heights (inches), allowing for tied observations.

Mothers x	62	63	64	64	65	66	68	70
Daughters y	64	65	61	69	67	68	71	65

Solution.

1. Assign average ranks. The two mothers of height 64 occupy ranks 3 and 4, so each receives 3.5. The two daughters of height 65 also occupy ranks 3 and 4 and receive 3.5.

Pair	R	T	R^2	T^2	RT
1	1	2	1	4	2
2	2	3.5	4	12.25	7
3	3.5	1	12.25	1	3.5
4	3.5	7	12.25	49	24.5
5	5	5	25	25	25
6	6	6	36	36	36
7	7	8	49	64	56
8	8	3.5	64	12.25	28
Total	36	36	203.5	203.5	182

2. Substitute the rank totals. Use $n = 8$, $\sum R = \sum T = 36$, $\sum R^2 = \sum T^2 = 203.5$, $\sum RT = 182$.

$$\rho_s = \frac{8(182) - 36(36)}{\sqrt{[8(203.5) - 36^2][8(203.5) - 36^2]}}$$

3. Simplify. The numerator is $1456 - 1296 = 160$; each bracket is $1628 - 1296 = 332$.

$$\rho_s = \frac{160}{\sqrt{332(332)}} = \frac{160}{332} = \boxed{0.48193}.$$

The association in ordering is positive and imperfect. Do not use the no-ties shortcut for this table.

fx-991CW – Manual ranks, direct result. Enter the paired rank columns, including 3.5, in 2-Variable. Check $n = 8$ and both means 4.5; read linear $r = 0.48192771$.

Exam point. Zero SD in either rank column makes rank correlation undefined. Average ranks plus Pearson works for both tied and untied data.

FOUNDATIONS / RELATIVE VARIATION AND GROWTH

Comparing spread and averaging change

Coefficient of variation

Compare SD with the mean, rather than comparing absolute SDs alone: Use CV $= 100\sigma/\mu\%$ or sample CV $= 100s/\bar{x}\%$. Smaller CV means greater relative consistency. Use a positive ratio-scale mean; CV is undefined at zero and can mislead near zero or on arbitrary-zero scales such as Celsius.

Illustration 27

Topics covered: Absolute versus relative dispersion; coefficient of variation; units; consistency; appropriate scales.

Problem. Factory A has mean output 20 units and population SD 2 units. Factory B has mean 100 units and SD 5 units. Compare their absolute and relative variation.

Solution.

1. **Absolute comparison.** A has lower SD, $2 < 5$, so less absolute spread in output units.
2. **Relative comparison.** Use CV $= 100\sigma/\mu$:

$$\mathrm{CV}_A = 100(2/20) = \boxed{10\%}, \qquad \mathrm{CV}_B = 100(5/100) = \boxed{5\%}.$$

B is relatively more consistent: its SD is a smaller percentage of its mean, despite being larger in absolute units.

fx-991CW – Calculator arithmetic. Type 100 × SD/mean. Choose sample or population SD consistently; CV is not a direct result and is undefined at mean zero.

Average compound growth

Each growth factor $= 1 +$ rate $(\%)/100$. For t years:

Annual rate $(\%) = 100[\sqrt[t]{\text{product of growth factors}} - 1]$.

The arithmetic average usually does not reproduce compounded growth.

Illustration 28

Topics covered: Percentage change; growth factors; geometric versus arithmetic mean; annual versus total growth.

Problem. A firm's net worth increases over five successive years by the percentages shown. Find (a) the total five-year increase, (b) the equivalent constant annual growth rate and (c) the arithmetic mean of the annual rates. Explain why (b) and (c) differ.

Year	1	2	3	4	5
Increase (%)	3	10.5	9	6	7.5

Solution.

1. **Multiply growth factors.** A rate $r\%$ multiplies the current value by $1 + r/100$.

$$M = (1.03)(1.105)(1.09)(1.06)(1.075) = 1.41364489825.$$

Thus total growth is $100(M - 1) = \boxed{41.36449\%}$.

2. **Match an equal annual multiplier.** With decimal annual rate g, $(1 + g)^5 = M$:

$$100g = 100(M^{1/5} - 1) = \boxed{7.16873\% \text{ per year}}.$$

3. **Average the reported rates arithmetically.**

$$\bar{r} = (3 + 10.5 + 9 + 6 + 7.5)/5 = \boxed{7.2\%}.$$

The geometric calculation reproduces successive compounding; the arithmetic average generally does not.

fx-991CW – Calculator arithmetic. Type $((1.03 \times 1.105 \times 1.09 \times 1.06 \times 1.075)^{(1/5)} - 1) \times 100$. Keep the full product until the final answer.

CASIO FX-991CW / ONE VARIABLE

Enter once, read the quantities you need

Source: Official Casio fx-570CW/fx-991CW guide, Statistics section; references on page 30.

Keys. HOME opens the apps. OK selects a menu item. EXE confirms an entry. TOOLS opens the context menu. Use menu names rather than memorising numbered choices.

Raw and frequency data

For a plain list select Statistics → 1-Variable with Frequency Off. For a count table set Frequency On *before* entry; changing this setting deletes existing Statistics data. The two problems show the key route and result checks.

Illustration 29

Topics covered: Raw data entry; count; sums; mean; population/sample variance and SD.

Problem. Using observations 2, 4, 6 and 8, obtain the count, sum, sum of squares, mean, population variance/SD and sample variance/SD on the fx-991CW. Show a brief written calculation to verify the two variances.

Solution.

1. Enter once. HOME → Statistics → 1-Variable; set Frequency Off before entry. Enter the four values in x and open TOOLS → 1-Var Results.

2. Verify totals and mean.

$$n = 4, \quad \sum x = 20, \quad \sum x^2 = 120, \quad \bar{x} = 20/4 = \boxed{5}.$$

3. Verify spread. Squared-deviation total $= 120 - 20^2/4 = 20$.

$$\sigma_x^2 = 20/4 = \boxed{5}, \quad \sigma_x = \sqrt{5} = \boxed{2.23607},$$

$$s_x^2 = 20/3 = \boxed{6.66667}, \quad s_x = \sqrt{20/3} = \boxed{2.58199}.$$

All requested quantities are direct outputs after entry. Select the population or sample pair from the question's wording; the displayed count n does not make that choice for you.

Illustration 30

Topics covered: Frequency entry; repeated values; count checks; sum of squares; weighted mean.

Problem. Use the calculator to find the total count, sum, sum of squares and mean for values 1, 2 and 3 with frequencies 2, 3 and 1. A student gets a count of 3 and mean of 2. Explain the error.

Solution.

1. Enter values and counts separately. Select 1-Variable and turn Frequency On before entry. Put $1, 2, 3$ in x and $2, 3, 1$ in Freq.

2. Verify the results by weighting.

$$n = 2 + 3 + 1 = \boxed{6}, \qquad \sum fx = 2(1) + 3(2) + 1(3) = \boxed{11},$$

$$\sum fx^2 = 2(1^2) + 3(2^2) + 1(3^2) = \boxed{23}, \quad \bar{x} = 11/6 = \boxed{1.83333}.$$

3. Diagnose the student's result. A count of 3 and mean $(1 + 2 + 3)/3 = 2$ indicate the frequencies were ignored: only the three distinct values were entered as equally frequent observations.

fx-991CW – Yes, directly. Read the count, two sums and mean in 1-Var Results. Correct frequency entry gives all four without expanding the six observations manually.

CASIO FX-991CW / GETTING THE MOST FROM IT

Paired data, ranks, moments and fast checks

A. Pearson correlation and sample covariance

HOME → Statistics → 2-Variable. Set Frequency as needed *before* entry. Enter paired x, y values in matching rows. Then use TOOLS → Reg Results → y=a+bx and read r. The linear choice matters; this is the Pearson correlation required here.

For sample covariance, obtain $n, \sum x, \sum y, \sum xy$ through Statistics Calc → CATALOG → Statistics, then evaluate

$$s_{xy} = \frac{\sum xy - (\sum x)(\sum y)/n}{n-1}.$$

fx-991CW – Calculator arithmetic. Insert the stored sums from CATALOG → Statistics in this expression. Sample covariance needs this calculation; r itself is a direct regression result.

The calculator's regression coefficient b is a slope, not covariance. For the table on page 19, expect $r = 0.5$ and $s_{xy} = 0.5$.

B. Spearman correlation: let Pearson process your ranks

Write the ranks on paper first, assigning average ranks to ties. Enter the *rank pairs* as two-variable data, choose linear Reg Results, and read r. That r is Spearman correlation of the original values. Entering raw values instead gives Pearson, not Spearman.

C. Skewness and kurtosis: efficient transformed-data entry

The CW does not give a one-step skewness or kurtosis coefficient in its Statistics results. Use its sums and powers:

1. Calculate the original mean and record it without premature rounding.
2. On paper calculate deviations $d = x - \bar{x}$ and their powers.
3. For $\sum fd^3$, enter d^3 in x with the original frequencies in Freq, then read $\sum x$. For $\sum fd^4$, enter d^4 similarly. **Record earlier results before replacing data.**
4. Divide each total by N to obtain m_3, m_4. Combine with m_2 using the formulas on page 17. Do not read SD of the transformed column as the original SD.

For the symmetric three-value example on page 17, only three distinct rows are needed; the cubed terms cancel, so you can skip an entire calculation.

D. Small arithmetic jobs that save time

For Chebyshev enter $(1 - 1/k^2) \times 100$; for a z-score enter $(x - \mu)/\sigma$; for a grouped quantile enter $L + ((pn - C)/f)h$. Use brackets around the full numerator and denominator.

For compound growth, enter the product of annual growth factors, raise it to 1/5, subtract 1, and multiply by 100. This is much faster than logarithms.

Exam point. Always check n, totals and denominator before trusting a result. The CW cannot choose a statistical model, justify causation, draw the required exam graph on paper, or decide your teacher's percentile convention. Worked solutions show what to write.

THEORY ANSWERS / WRITE IN EXAM LANGUAGE

Short answers with the reasoning included

1. Why is dispersion needed if we already know the mean?

The mean gives a central value but does not describe how closely observations cluster around it. Two distributions can have the same mean and different variation. Range, IQR, variance and SD describe that variation. Lower dispersion means greater consistency around the centre, as illustrated by the two delivery-time lists.

2. Distinguish variance and standard deviation

Variance is the average squared deviation from the mean for a population; the usual sample variance uses $n-1$ to estimate population variance. SD is the positive square root of variance. Variance has squared units, while SD has the original data units and is easier to interpret on that scale.

3. State Chebyshev's theorem and its use

For a distribution with a finite mean and positive finite SD, at least $1-1/k^2$ of values lie within k SDs of the mean, for $k>1$. It provides a minimum proportion without assuming normality. In particular, at least 75% lie within two SDs. It does not give an exact probability.

4. Distinguish skewness and kurtosis

Skewness describes asymmetry and its direction. Positive skewness is associated with a longer right tail; negative skewness with a longer left tail. Kurtosis describes the influence of extreme deviations, traditionally introduced as peakedness. Pearson kurtosis β_2 is compared with the normal benchmark 3 to classify leptokurtic, mesokurtic and platykurtic shapes.

5. Why does correlation not imply causation?

Correlation describes an observed association. It does not by itself identify why that association occurs. A third variable may affect both measurements, the direction of influence may be reversed, or the association may be accidental or due to selection.

For example, ice-cream sales and swimming incidents may both rise in hot weather. Their positive correlation does not show that buying ice cream causes an incident. Weather is a plausible common cause. A causal conclusion needs suitable design and evidence beyond the coefficient alone.

Illustration 31

Topics covered: Association; common causes; reverse causation; evidence needed for a causal claim.

Problem. A town finds a positive correlation between ice-cream sales and swimming incidents. A school finds that students with more tutoring hours have lower marks. Does either association alone establish causation? Give a plausible explanation for each.

Solution.

1. Town example: a common cause. Hot weather can increase both ice-cream purchases and swimming activity. More swimming exposure can accompany more incidents. The observed correlation does not show that buying ice cream causes an incident.

2. School example: possible reverse direction. Students who are struggling may seek more tutoring. Low performance can lead to more tutoring, rather than tutoring causing lower performance.

3. Conclusion. Neither association alone establishes a cause. A causal claim needs suitable study design and evidence that addresses alternative explanations.

fx-991CW – Interpretation is manual. The CW can compute r from paired numeric data. It cannot identify confounders or establish causal direction. No data table or coefficient was supplied here, so no numerical r can be calculated.

6. Which visualization should I use?

Task	Suitable display and what to label
One numeric variable	Histogram: class boundaries on horizontal axis, frequency on vertical axis; adjacent bars touch
Compare spread	Boxplot: Q_1, median, Q_3; state the whisker rule if used
Two numeric variables	Scatterplot: one point (x, y) per pair; label both axes and units
Compare categories	Bar chart: category labels and counts; separate the bars

Illustration 32

Topics covered: Histogram; class boundaries/frequencies; boxplot; bar chart; scatterplot; axes and pairing.

Problem. For scores 5, 5, 15, 15, 15, 15, 25 and 25, construct a frequency table for continuous classes 0–10, 10–20 and 20–30, and draw a histogram and boxplot. Use median-of-halves quartiles and min/max whiskers. Also name the display for paired hours–marks and category counts.

Solution.

1. Count by class.

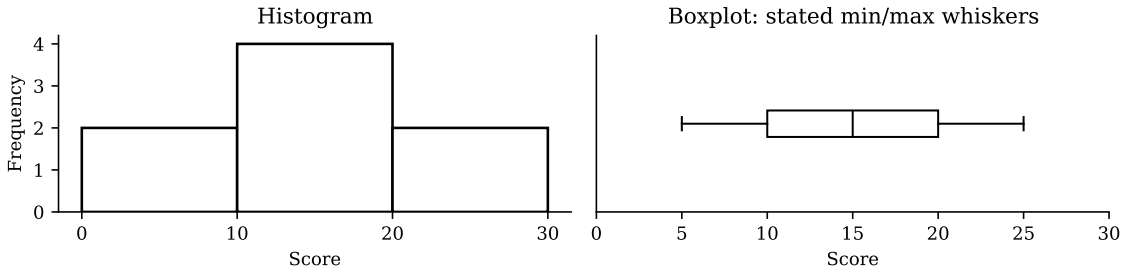
Class	0–10	10–20	20–30
Frequency	2	4	2

The histogram has touching bars with these heights and class boundaries on its horizontal axis.

2. Calculate boxplot positions. The middle two values are both 15, so median = 15. Lower-half middle values are 5 and 15; upper-half middle values are 15 and 25:

$$Q_1 = (5 + 15)/2 = \boxed{10}, \quad Q_2 = \boxed{15}, \quad Q_3 = (15 + 25)/2 = \boxed{20}.$$

Draw the box from 10 to 20 with a median line at 15; the specified whiskers end at minimum 5 and maximum 25.



3. Select other plots. Use a scatterplot for matched hours–marks and separated bars for category counts.

fx-991CW – Summaries direct, drawing manual. Read raw median/quartiles when conventions match, then draw the labelled graphs yourself. The CW does not display a full histogram or scatterplot graph.

Exam point. For Pearson, describe *linear* association. A zero coefficient does not rule out a curved relationship. "Strong" labels are context-dependent; support the description with the actual coefficient and plot.

LAST-PASS REVISION / 1 OF 2

Centre, spread and positions

Quantity	Formula / decision
Mean	$\mu = \sum x/N$; $\bar{x} = \sum x/n$; with counts, replace sums by $\sum fx$ and size by $\sum f$ **Calculator:** Direct: 1-Var Results, $\bar{x}$; Frequency On for counts.
Range / IQR	$R = x_{\max} - x_{\min}$; IQR $= Q_3 - Q_1$; QD $=$ IQR$/2$ **Calculator:** Read min/max or quartiles, then subtract; divide IQR by 2 for QD.
Population variance	$\sigma^2 = \sum(x-\mu)^2/N = \sum x^2/N - \mu^2$ **Calculator:** Direct: 1-Var Results, σ_x^2.
Sample variance	$s^2 = \sum(x-\bar{x})^2/(n-1) = [\sum x^2 - (\sum x)^2/n]/(n-1)$ **Calculator:** Direct: 1-Var Results, s_x^2.
Standard deviation	$\sigma = \sqrt{\sigma^2}$ or $s = \sqrt{s^2}$; original units **Calculator:** Direct: read σ_x (population) or s_x (sample).
Frequency data	Multiply every contribution by f; use midpoints for class intervals **Calculator:** Direct after entering values or midpoints in x, counts in Freq.
Grouped quantile	$Q_p = L + [(pn - C)/f]h$; C is cumulative count before the required class **Calculator:** Choose the class manually; type the interpolated formula.
Grouped mode	$L + [(f_1 - f_0)/(2f_1 - f_0 - f_2)]h$ for equal-width classes **Calculator:** Choose modal class manually; type the formula. No direct mode output.
CV	$100\sigma/\mu\%$ or $100s/\bar{x}\%$; smaller means greater relative consistency **Calculator:** Type 100 × SD ÷ mean; no direct CV result.
Chebyshev	At least $1 - 1/k^2$ in $\mu \pm k\sigma$, $k > 1$ **Calculator:** Type $(1 - 1/k^2) \times 100$; calculate endpoints separately.
Empirical / z	Normal approximation: 68–95–99.7%; $z = (x - \mu)/\sigma$ **Calculator:** Remember the percentages; type $(x - \mu)/\sigma$ for z.

Choose the calculator output before calculating

Question wording	Read or calculate
Entire population	Read σ_x^2 for variance, σ_x for SD
Sample estimating population spread	Read s_x^2 for variance, s_x for SD
Frequency table	Turn Frequency On before entry; counts are not class totals to average
Grouped classes	Enter midpoints/counts for mean and SD; interpolate quartiles manually

Exam point. A sample mean divides by n. The $n-1$ correction belongs to sample variance, not the mean. Always keep the calculator's full precision until the final answer.

LAST-PASS REVISION / 2 OF 2

Shape, correlation and final checks

Quantity	Formula / decision
Pearson skewness	$(\bar{x} - \text{Mode})/s$ or $3(\bar{x} - \text{Median})/s$ **Calculator:** Type the ratio using mean/SD results and mode or Med; no direct skewness.
Bowley / Kelly	$(Q_3 + Q_1 - 2Q_2)/(Q_3 - Q_1)$; $(P_{90} + P_{10} - 2P_{50})/(P_{90} - P_{10})$ **Calculator:** Use quartile outputs or manually found percentiles, then type the ratio.
Moments	$m_2 = \sum f(x-\mu)^2/N$, $m_3 = \sum f(x-\mu)^3/N$, $m_4 = \sum f(x-\mu)^4/N$; $\gamma_1 = m_3/(m_2\sqrt{m_2})$, $\beta_1 = \gamma_1^2$ **Calculator:** $m_2 = \sigma_x^2$ is direct; form and sum powers for m_3, m_4.
Kurtosis	$\beta_2 = m_4/m_2^2$; excess $\gamma_2 = \beta_2 - 3$; compare with 3 or 0 respectively **Calculator:** Type m_4/m_2^2 and subtract 3; no direct kurtosis result.
Covariance	Sample $s_{xy} = \dfrac{\sum xy - (\sum x)(\sum y)/n}{n-1}$ **Calculator:** Use 2-Variable sums in this formula; no separate covariance output.
Pearson correlation	$r = \dfrac{n\sum xy - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$ **Calculator:** Direct: 2-Variable, Reg Results, y=a+bx, read r.
Spearman	No ties: $1 - 6\sum d^2/[n(n^2-1)]$; ties: Pearson on average ranks **Calculator:** Rank manually, then linear Pearson r on the paired ranks.
Compound growth	Annual growth (%) $= 100[\sqrt[t]{\text{product of growth factors}} - 1]$; each factor $= 1 + \text{rate}/100$; t counts years **Calculator:** Type product of factors, raise to $1/t$, subtract 1, multiply by 100.

Before submitting

1. Name the quantity and write the formula with the correct denominator.
2. Show the data substitution or working-table totals.
3. For correlation, preserve the pairs and choose original values for Pearson or ranks for Spearman.
4. Give units where needed and a sentence interpreting the result.

Exam point. Correlation describes association. Pearson describes a linear pattern; Spearman describes ordering. Neither proves causation. Zero Pearson correlation does not rule out a curved relationship.

Sources and conventions. Aligned with the supplied *Unit 3–Descriptive Statistics–Dispersion*, *Unit 3–Skewness–Kurtosis* and syllabus image; presentation follows the supplied *Statistics Exam Guide*. Illustrations combine foundation exercises with representative applications drawn from the supplied notes and correlation sheet.

Calculator reference: Casio, *fx-570CW/fx-991CW User's Guide*, Statistics section: https://support.casio.com/en/manual/004/fx-570CW_991CW_EN.pdf. Supplemental coefficient definitions: IGNOU, *BECC-107*, Block 1, Unit 4, https://egyankosh.ac.in/bitstream/123456789/73726/1/Block-1.pdf. Raw-quantile and moment conventions are stated in the relevant theory.